

A Novel Framework to Facilitate User Preferred Tuning for a Robotic Knee Prosthesis

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Abstract—The tuning of robotic prosthesis control is essential to provide personalized assistance to individual prosthesis users. Emerging automatic tuning algorithms have shown promise to ease the device personalization procedure. However, very few automatic tuning algorithms consider the user preference as the tuning goal, which may limit the adoptability of the robotic prosthesis. In this study, we propose and evaluate a novel prosthesis control tuning framework for a robotic knee prosthesis, which could enable user preferred robot behavior in the device tuning process. The framework consists of 1) a User-Controlled Interface that allows the user to select their preferred knee kinematics in gait and 2) a reinforcement learning-based algorithm for tuning high-dimension prosthesis control parameters to meet the desired knee kinematics. We evaluated the performance of the framework along with usability of the developed user interface. In addition, we used the developed framework to investigate whether amputee users can exhibit a preference between different profiles during walking and whether they can differentiate between their preferred profile and other profiles when blinded. The results showed effectiveness of our developed framework in tuning 12 robotic knee prosthesis control parameters while meeting the user-selected knee kinematics. A blinded comparative study showed that users can accurately and consistently identify their preferred prosthetic control knee profile. Further, we preliminarily examined gait biomechanics of the prosthesis users when walking with different prosthesis control and did not find clear difference between walking with preferred prosthesis control and when walking with normative gait control parameters. This study may inform future translation of this novel prosthesis tuning framework for home or clinical use.

Index Terms—Robotic knee prosthesis, human-in-the-loop optimization, user preference, reinforcement learning, user-controlled interface.

I. INTRODUCTION

LOWER limb amputation causes permanent disability, impacting patients physically, psychologically, and socially [1]. Advancing prosthetic technology is needed to enhance individuals' mobility and quality of life with lower limb amputation. One promising technology is robotic prosthetic legs that are motorized and intelligently controlled. Compared to passive alternatives that fail to replicate the behaviors of the healthy limb across various environments, robotic prostheses showed many potential advantages [2], including increased range of motion [3], enhanced stability and balance of wearers [4], [5], and ability to adapt to various terrains [6], [7].

A frequently used control approach for robotic knee devices is a finite-state impedance control (FSM-IC), which can reasonably simulate human limb behavior without complex implementation [8]. Compared to trajectory based control of robotic prosthesis devices FSM-IC provides more compliance with the user along with flexibility during the operation [9], [10]. However, personalization of the controller's impedance parameters (often over 9) is usually needed and has been one of the main challenges for FSM-controlled prosthesis devices in practice [11]. In the clinic, prosthetists will typically manually tune the parameters to obtain optimal prosthesis control performance in assisting the user's gait [12]. This procedure is highly subjective, time inefficient, and non-standardized.

To overcome the limitations of the manual tuning process of control parameters, an autotuning approach using reinforcement learning (RL) has been proposed by our group for robotic lower limb prostheses previously [13], [14]. The RL-based automatic control method optimally tuned the impedance parameters of each gait phase to replicate the normative profile (NP) of knee kinematics gained from the non-disabled population during gait [15]. Further, this approach has been improved in terms of time efficiency and robustness of the learned tuning policy [16]. The algorithm had been successfully validated with transfemoral (TF) amputees using the robotic knee prosthesis while ensuring their safety. Although promising, one open question is whether reproducing the

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normative profile of knee motion should be the tuning goal that can satisfy the prosthesis user's preference. Potentially, physiological changes in amputees due to limb amputation and the dynamic difference between the robotic prosthesis and biological limb may lead to different preferred knee kinematics rather than NP [17]. Thus, consideration of the user preference might be important for personalizing the prosthesis control. Studies showed that the lack of user input during the device selection, fitting time frame, and changing user needs are among the main reasons for the prosthesis abandonment [18], [19]. In addition, prosthetists make prosthesis control tuning decisions based on not only the observational gait analysis but also the user's preference and feedback in clinics [20], [21]. These insights signify the importance of prosthesis user's preference consideration in regard to prosthetic behavior.

Addressing user preference/satisfaction in wearable robot control optimization has recently become an emerging topic in the research community. Predictive Entropy Search with Preferences algorithm has been developed by Thatté et al. [22] in which user preference is incorporated by pairwise comparisons of different controls. Tucker et al. [23] developed LineCoSpar to learn the utility function via a pairwise comparison of user preferences. The idea was to learn six user-preferred gait parameters of the lower body exoskeleton by considering the user's feedback for each posterior update. These approaches have finite forced choices from which users select their most preferred behavior. Quesada et al. [24] studied the effect of ankle prosthesis push-off work on the metabolic cost and user satisfaction. Transtibial amputees who participated in the study demonstrated strong preferences over various levels of ankle push-off work despite not experiencing decreased metabolic rates. More recently, approaches where users tune quasi-passive ankle prosthesis stiffness [25] or torque profile of bilateral ankle exoskeleton [26] to find their preferred device control parameters have been successfully demonstrated. These preference-based approaches have been built upon successfully demonstrating perception of users to changes in control parameters of their prosthetic and wearable devices [27], [28], [29]. These existing studies enabled the researchers to understand the importance of users' preferences in the wearable robot control and also motivated us to include user preference in an FSM-IC control to further leverage the FSM controller benefits. However, one challenge facing us is that the FSM-IC control includes 12 impedance control parameters [30], while all the existing, related studies only explored 1 to a maximum of 6 control/action parameters in the wearable devices. Exploring this high-dimensional device control space by the humans for maximal user preference is difficult, if possible, and has never been demonstrated based on our knowledge. Other solutions that can reduce human adjustable parameter space for FSM-IC controlled device is needed.

To address this challenge and enable the tuning of 12 knee prosthesis control parameters to meet the user's preference while ensuring user safety and minimizing human exertion, we proposed a novel hierarchical engineering framework

TABLE I
DETAILED INFORMATION OF THE SUBJECTS

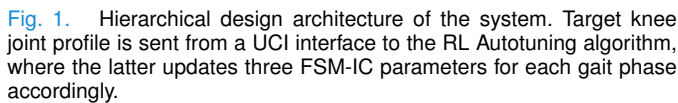
Subject	Sex	Prosthesis Side	Height (m)	Weight (kg)	Age
AB01	Female	Right	1.86	81.6	23
AB02	Male	Left	1.87	68	21
AB03	Male	Left	1.93	70.3	25
AB04	Male	Left	1.91	75.9	28
AB05	Female	Left	1.90	77.1	20
TF01	Male	Left	1.74	80	23
TF02	Male	Left	1.74	68	42
TF03	Male	Left	1.70	68	67
TF04	Male	Left	1.81	89	38
TF05	Male	Left	1.79	72.5	22

consisting of (1) our previously developed RL-based prosthesis tuning method [16] and (2) a User Controlled Interface (UCI) [31]. The basic working mechanism was that the human users can use the high-level UCI to determine their own preferred prosthesis knee kinematic features in a 4-dimensional space by modifying control points which is then realized by the low-level RL-based tuning algorithm adjusting all 12 impedance control parameters to meet the desired kinematic features. Since our RL tuning algorithm was time-efficient and the learned prosthesis tuning policy was robust across various kinematic profiles, this framework offered us a unique opportunity to investigate the user's preferences across various knee prosthesis control parameters. Our pilot testing showed the design and feasibility of this engineering framework in adjusting stance phase knee features [31].

In this study, we implemented and evaluated the framework that can tune 12 prosthesis control impedance parameters to meet the user defined features in knee kinematics across gait cycle. Since the perception of users to changes in robot behavior is essential for these users to have a preference, we investigated if users can differentiate between the profile they chose, the normative profile and a random profile to showcase the need for the current framework. Finally, we investigated some potential biomechanical factors that could drive users to choose one profile over the other. The results of this study may lead to a novel wearable robot tuning framework that can personalize high-dimensional device control parameters to meet the user's preference in the future.

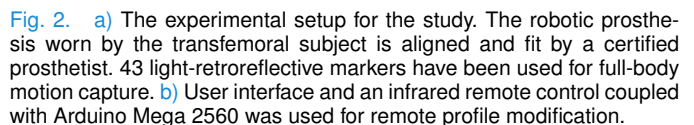
II. METHODS

To achieve the aims of the study, the experiment was designed to evaluate the performance of the framework, by helping users chose a preferred profile and perform comparison studies to analyze users' ability to differentiate different knee profiles. Ten participants, five non-disabled (AB) and five with unilateral above-knee amputation (TF) participated in the study, with demographics as listed in Table I. All amputee participants were community ambulators (K3 activity level or higher) without significant secondary comorbidities, who regularly used conventional passive prostheses and were active in daily life. The study was conducted with the approval of the Institutional Review Board of the University of North Carolina



A. Robotic Knee Prosthesis and User-Controlled Interface (UCI)

Trained RL policy adjusts three FSM-IC parameters (stiffness, equilibrium angle, and damping) defined for each gait phase according to the differences between measured features and target features in the knee profile, referred to as the peak error and duration error [16]. The convergence criteria for the autotuning algorithm were to have the four control points (CP1 – CP4) of the knee profile which were appropriate to prementioned gait phases to be within 2 degrees spatially of the target control points and 2% temporally for 6 out of 10 consecutive impedance updates. We integrated the User-Controlled Interface (UCI) into the autotuning architecture (Fig. 1.) that enabled users to interact with the robotic prosthesis by having them set their preferred kinematics for the knee joint. Users were given a remote controller to modify (up and down) the predefined magnitude of four control points corresponding to gait phases shown in Fig. 2(b). The RL



B. Experimental Setup

C. Experimental Protocol

The experiment duration was two consecutive days for each participant and was divided into the following sessions: *profile exploration* and *pairwise blinded profile comparison sessions*.

In the first session, the users were fitted with the prosthesis, which was then tuned to simulate a normative knee kinematic profile (NP). Then to familiarize the participants with how changing the control point location affects the robot behavior, we let the participants explore different profiles in which one of the four control point was changed while the rest were set to NP. They were first instructed to move the first control point above its NP location by 3 degrees and the resulting profile was tuned. Then they are instructed to move the control point below its NP location by 3 degrees and the tuning is performed. The process is then repeated for the remaining three control points resulting in tuning of 8 profiles, two for each control point. It should be noted that each control point has been tuned in isolation from the other three. To reduce fatigue, a 2-minute rest was provided for each participant after every 2-minute interval of walking on the treadmill for tuning trials.

Upon completion of the tuning of the nine profiles (one NP and eight modified profiles), subjects were then instructed to experience each of the profiles so that they understand how moving the control point affects robot behavior. This was achieved by instructing the participants to walk on the treadmill with the powered prosthesis replicating NP, $+3^\circ$ and -3° profile for each control point for 45 seconds. As a result, each participant walked in twelve 45 seconds of treadmill walking bouts where NP profile repeated four times. At the end, the subjects were also asked which of the three profiles they prefer for each control point, and their preference was noted. Thus, at the end of these search trials, we were able to construct a starting point for preferred knee joint profile according to participants' preferences regarding each phase.

For the second session, the robotic prosthesis was fit to the user and was tuned again to replicate NP to account for differences in socket fitting. Then the prosthesis was tuned to the constructed profile acquired using the preferred control points' locations from the previous day. This profile is constructed by joining the selected control points using minimum jerk profiles. Once tuned, the participant walked with this profile to verify if they were satisfied with it or if any adjustment was needed. Any adjustments were then incorporated, and the resulting profile was tuned. Once the participant was satisfied with the profile, the control parameters were saved as the person's preferred profile (PREF). Finally, in order to eliminate the random choice effects during the blinded comparison session (detailed below), we tuned a profile called opposite profile (OPP). OPP profile mirrored the preferred profile with respect to the NP, e.g., if a participant chose $+3^\circ$ for a CP1, then the OPP profile is tuned to -3° at that control point and vice versa.

Once the three profiles were tuned, 12 pairwise blinded comparisons (4 sets of each of the three pairs: NP vs. PREF, NP vs. OPP, PREF vs. OPP) were held at the end of the second session. The goal was to investigate if subjects can actually perceive the differences between profiles and have a preference when they are blinded to the profile they are experiencing. To minimize the order effect, the order of the pairwise comparisons and the order of the trials in the pairwise comparisons were randomized for each participant.

Each participant was asked to walk with both profiles in a pair for 45 seconds without knowing which two profiles out of three they were experiencing. After experiencing 45 seconds of walking for each profile, participants were asked about which of the two profiles they preferred for walking and if they could recognize them. The consistency of preference and user's ability to recognize profiles was then analyzed. They were allowed to skip the recognition of any profile if it was truly hard for them to recognize (NP, PREF, or OPP).

D. Data Analysis and Evaluation Metrics

The proposed autotuning framework was evaluated by quantitative and qualitative aspects. First, we quantified the performance of the RL-learned tuning policy by (1) the success rate in meeting targeted knee kinematics (i.e., effectiveness) and (2) the duration needed to meet targets (i.e., time efficiency) after the targeted knee profile was modified. In addition, we quantified UCI usability by a qualitative survey with items adapted from the System Usability Scale [34], consisting of 9 questions and the space for general feedback.

To quantify the consistency of participants' perceptions about their chosen profile, the percentage of chosen profiles during the blinded pairwise comparison was calculated. The aim was to provide insights about the choices of profiles over each other (NP vs. PREF, NP vs. OPP, PREF vs. OPP). In addition, we quantified the number of times when participants were able to recognize or mislabel the experienced profiles.

In addition to exploring the user's perception on various profiles, some possible biomechanical factors that could influence the subjects' preference for a profile were analyzed. To that effect, temporal and step length symmetry [35], and margin of stability [36] were analyzed. Joint kinematics and kinetics were calculated using the conventional inverse dynamics approach with the Plugin-Gait software (PiG, Vicon, Oxford, UK) [15], [37], [38], [39]. MATLAB 2020a was used to obtain a symmetry and stability index for further data analysis. Thirty-five seconds of steady-state gait data from the pairwise comparisons were analyzed for each profile to analyze these factors. A fourth-order Butterworth filter smoothed GRF signals with a cutoff frequency of 25 Hz to reduce noise while the marker positions were low-pass filtered by a fourth-order Butterworth filter with a cutoff frequency of 10 Hz. The data was then split into gait cycles using the GRF of each limb with a threshold of 30 N. The average gait cycle data was then evaluated using (1) to evaluate temporal and step length symmetry.

$$SI = \frac{x_i - x_p}{(x_i + x_p) * 0.5} * 100\% \quad (1)$$

where SI – symmetry index, x_i and x_p are intact and prosthesis side parameter (i.e., step length or stance time in this study). The value of $SI = 0$ indicates perfect symmetry. Dynamic margins of stability measures were adapted from Hof et al. [40] based on the extrapolated center of mass (XcoM) equation

shown below:

$$X_{coM} = y + \frac{\dot{y}}{\omega_0} \quad (2)$$

$$\omega_0 = \sqrt{\frac{g}{l}} \quad (3)$$

where y is the COM position in sagittal plane, $\dot{y}$ is the COM velocity and ω_0 is the angular frequency of the approximated inverted pendulum estimated through (3), in which g is the acceleration due to gravity, and l is the equivalent pendulum length, which was taken 1.34 times the leg length in this study [41]. The dynamic margin of stability (MOS) was then defined as:

$$MOS = X_{coM} - BOS \quad (4)$$

where BOS was the boundary of the base of support (i.e., the most anterior edge of the leading foot). We further defined the anterior-posterior (AP) margin of stability (MOS_{AP}) as the AP distance between the X_{coM} and the toe marker of the leading foot.

To determine significant differences among compared biomechanics metrics, interparticipant mean and standard error for each compared knee joint profile were calculated across all participants and the Friedman test was performed accordingly.

III. RESULTS

A. Autotuning System Performance

Based on the criteria set for control tuning convergence mentioned in the methods, the RL-learned tuning policy could achieve 100% successful rate to tune the prosthesis knee control parameters to meet various target profiles chosen by different subjects, showcasing the robustness of the algorithm. The time duration required to tune all tested prosthesis profiles among all participants was 2.5 ± 1.4 minutes. That means after each prosthesis targeted profile change (via UCI), it took around 2-3 minutes of walking for RL to tune prosthesis control to reach the new targeted knee motion. Fig. 3. (a) shows the representative convergent behavior of the RL autotuning algorithm for NP profile tuned for TF05 whereas Fig. 3. (b) demonstrates the number of iterations needed to tune all three profiles for all participants. It should be noted that each iteration constitutes to four full gait cycles.

Table II summarizes the response of the participants to questions asked for the evaluation of the UCI usability. The number in each box denotes the number of participants who gave the rating to the specific question. 7 out of 10 participants agree that the UCI was easy to use and could be learned to use very quickly.

B. User Preferred Knee Profiles

Fig. 4. shows the preferred profiles for each of the participants along with the normative profile as a comparison. Eight out of ten participants preferred the CP1 to be different from that of NP, with 7 of them choosing a lower knee angle value. In comparison, 9 out of 10 participants chose a different location for CP2, while only 5 and 7 subjects chose a different profile for CP3 and CP4 respectively. Looking

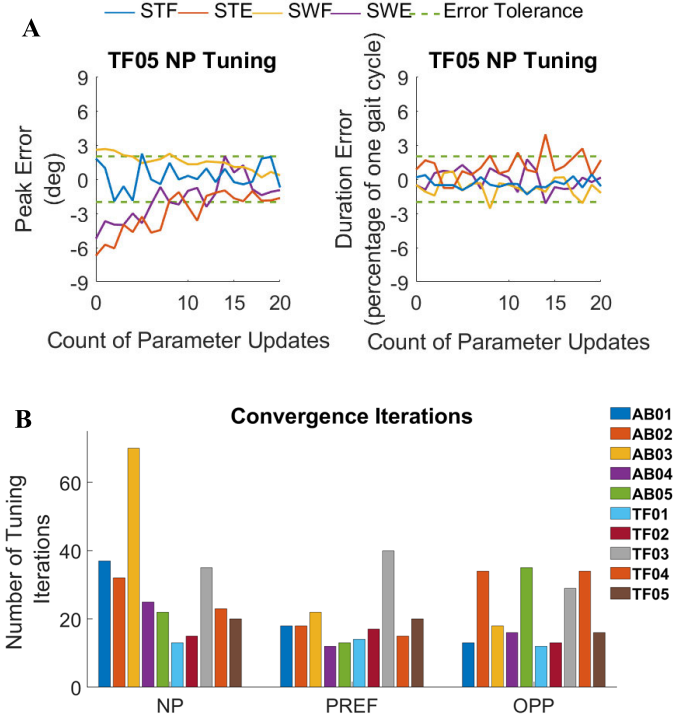


Fig. 3. a) Representative figure showing the spatial and temporal convergence of the RL algorithm to tune the 12 control parameters of robotic prosthesis according to NP kinematics provided. b) Number of tuning iterations needed to achieve the full convergence over three profiles and all participants. One tuning iteration constitutes to four gait cycles.

TABLE II
UCI USABILITY SURVEY RESULTS

	Strongly Disagree			Strongly Agree		
1. I found the UCI unnecessarily complex.	7	3	-	-	-	-
2. I thought the UCI was easy to use.	-	-	2	2	6	-
3. I think that I would need assistance to be able to use the UCI.	3	3	2	1	1	-
4. I found the various functions in the UCI were well integrated.	-	1	1	7	1	-
5. I thought there was too much inconsistency in the interface.	5	4	1	-	-	-
6. I would imagine that most people would learn to use this interface very quickly.	-	2	-	5	3	-
7. I found this interface very cumbersome/awkward to use.	4	4	1	1	-	-
8. I felt very confident using this interface.	-	-	1	6	3	-
9. I needed to learn a lot of things before I could get going with this interface.	6	2	2	-	-	-

into the preference, six participants (4 TFs) preferred to increase the second control point (CP2) whereas three of them (1 TF) decided to decrease it. For the last control point, there were no particular trend. On an individual basis, four participants (AB02, AB03, TF03 and TF05) preferred to achieve a relatively flat trajectory for the stance phase with

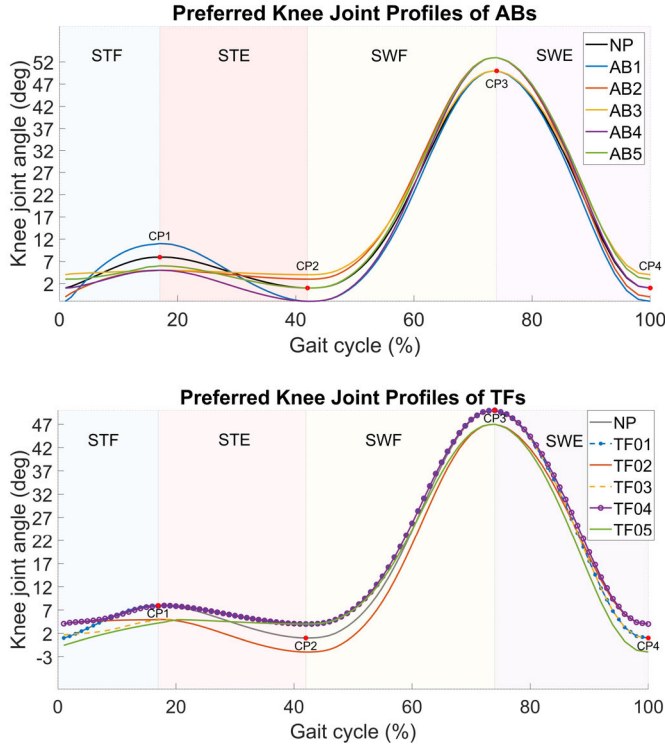


Fig. 4. Preferred knee joint profiles for non-disabled (AB) and amputee (TF) participants are shown along with normal knee joint profile. Control points are shown with red dots. Plots are divided into four gait phases: stance flexion (STF), stance extension (STE), swing flexion (SWF), and swing extension (SWE).

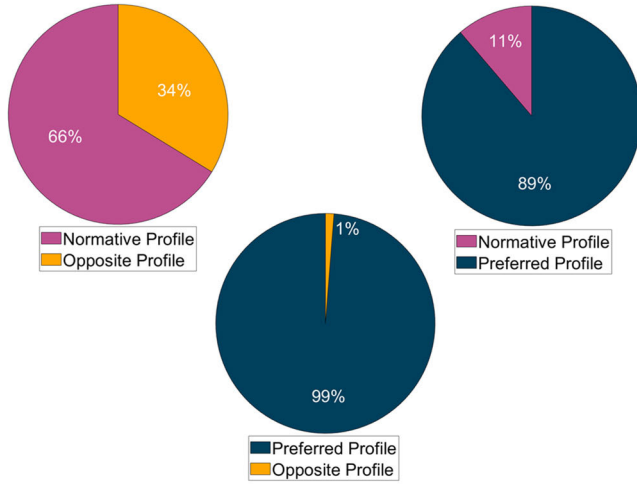


Fig. 5. Pairwise profile choice results show the percentage of each chosen profile for all 10 participants. NP has shown with pink, PREF with blue and OPP with yellow color.

similar values for CP1 and CP2. Additionally, four participants (AB03, AB05, TF02 and TF04) preferred to increase the CP4 while decreasing the CP1. Overall, there was no particular profile preference trend observed among all the participants.

C. Pairwise Comparison Analysis

In accordance with the study's aims, once the preferred profile was obtained, the participants' ability to consistently

TABLE III
PROFILE RECOGNITION CONFUSION MATRIX

Recognized Profiles		True Profiles		
		NP	PREF	OPP
	NP	38.8%	18.8%	28.8%
	PREF	16.3%	73.8%	5%
	OPP	41.3%	3.8%	63.8%

perceive different profiles and recognize their preferred profile was evaluated. The distribution of participants' choices upon pairwise comparison of their preferred profile, NP and OPP profiles is shown in Fig. 5. Participants chose the preferred profile over the opposite profile almost in all comparison trials. Participants also chose the preferred profile over NP 89% of the time. The choice rate of NP over OPP profile was 66%. The comparison results suggests that users can physiologically differentiate between the profiles and have distinct preferences.

A confusion matrix based on profile recognition data combining all the subject responses was developed and shown in Table III. It should be noted that the cases that participants could not recognize the experienced profile are excluded from the calculation of the confusion matrix. Since each subject experienced each individual profile (NP, PREF and OPP) 8 times during the comparison tests, there were a total of 80 datapoints. Participants were able to recognize the preferred profile with a 73.8% recognition rate. Around 18.8% of the tested preferred profile was mislabeled as NP, and only 3.8% of the tested preferred profile was recognized as the opposite profile. The opposite profile was correctly recognized in about 63.8% of cases. Most of the confusion errors occurred between the opposite profile and NP. The comparison between Fig. 5. and Table III shows that even in cases where the profile wasn't recognized, there was a preference for PREF profile over other profiles.

D. Biomechanical Analysis

Lastly, we explored gait symmetry and balance index that may potentially be correlated with the users' preferences in prosthesis control. Only TF data is reported here.

The interparticipant mean and the standard error (SE) across participants for the stance time and step length symmetry indexes across three knee joint profiles is shown in Fig. 6. along with similar measurements for the margin of stability in anterior-posterior direction. The Friedman test across 5 TF participants over 8 trials for each knee joint kinematics showed no significant differences for all three indexes utilized (Fig. 6.).

IV. DISCUSSION

In this study, we demonstrated the function of UCI, hierarchically combined with the RL-based prosthesis tuning

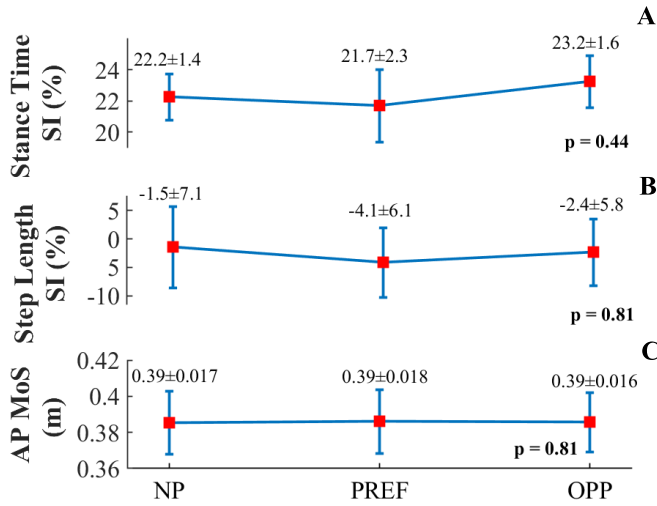


Fig. 6. Red squares and blue error bars represent the TF inter-participant mean and standard error (over eight trials) for each knee joint profile for (a) stance time (b), step length symmetry indexes, and (c) margin of stability in the anterior-posterior direction. The p-value for each compared metric is shown at the top of each graph.

algorithm, to tune 12 prosthesis control parameters that can be potentially used to tune the kinematic behavior of the robot based on user's preference. In addition, the study showed that users can perceive different profile and given a choice, their chosen profile is generally different from a normative profile, implying the need to consider user preference. Further, understanding the user's decision-making process in selecting preferred knee kinematics features and the gait biomechanics associated with the user's preference could provide additional insights to the future development of controller and tuning algorithms that align with user preferences, thereby leading to better technology adoption. Finally, some preliminary biomechanical analysis has been performed to investigate if there is a specific factor that leads to users preferring one profile over the other.

A. RL Based Autotuning Algorithm and UCI Framework

According to the results of the study, the RL-based autotuning algorithm demonstrated its time efficiency and robustness of the learned prosthesis tuning policy across various kinematic profiles. Without re-updating the tuning policy, when the knee kinematics profile changed, the algorithm could tune the 12 control parameters in under 40 iterations for most subjects safely to reach the new target profile. The ability of the algorithm to tune over the high-dimensional device control space suggests the possible feasibility of the approach being deployed outside of the laboratory settings to tune prosthesis control parameters according to target knee joint kinematics. In addition, participants' responses to the UCI usability survey showed that the interface was easy to use, and the users were confident when using the interface. However, the participants required additional explanation of the meaning of the knee kinematics associated with their walking. This could be addressed by developing a detailed video to explain gait and knee features in layman's terms prior to their UCI based tuning session.

B. Preferred Profile Analysis and User Preference

Results showed that the preferred knee motion curves were unique for each participant. While replicating NP using a robotic prosthesis has been the current predominant approach, none of the participants preferred the NP. Based on the profiles, while some participants preferred a more rigid prosthetic during the stance phase, others preferred a more compliant approach by having higher flexion of the knee before the heel strike that was immediately changed to a more rigid extension. However, no explicit assertions were observed in user feedback, although some users mentioned stability and lack of wobble as their driving factors, which could explain the preference of stiffer prosthesis. This observation implied that users had a specific preference even if they might not verbalize the specific qualities that they were looking for. It also highlighted the importance of having an approach that can let the prosthesis users explore different possibilities of prosthetic behavior. In addition, the profile exploration trials for this study have been performed sequentially and there was no way for participants to assess how changing one control point changes their perception of the others. Thus, our ongoing study aims to analyze the psychological aspects that can drive prosthesis users' choices during the extensive exploration of their preferences while also considering the interdependent relationships between control points that could affect profile perception [42].

In most cases, participants successfully chose their preferred knee profiles during the blind pairwise comparison session, supporting the assumption that participants were able to physiologically differentiate between the experienced profiles. In addition, participants were able to recognize their preferred knee profile with moderate accuracy, even with limited exposure. This observation implies that clinicians and robotic prosthesis manufacturers could rely on the consistency of users' opinions for tuning the prosthetic.

Given the effectiveness and time efficiency of our proposed framework to tune prosthesis control based on user's preference and the consistency in the user's perceived preferences, our framework may be potentially used by the prosthetists in clinics and even be transferred to home for prosthesis users. In addition, all the sensing in the tuning algorithm being internal to the device makes it a possible solution for clinics and home environment because extensive gait analysis instrumentation tools are not easily available in such healthcare facilities. However, practical challenges related to the hardware and safety need to be addressed before the system can be implemented at home. One of our future works is to let prosthetists use the software and provide feedback on our design.

C. Biomechanical Analysis

Effects of user preferred prosthesis control on gait performance metrics related to gait symmetry and stability were preliminarily examined. The main gait performance metrics chosen for analysis were symmetry (stance time and step length) and balance (margin of stability). These factors were chosen based on their impact on the amputee users. It has shown that gait asymmetry causes long-term health conditions [43]. Further, the loss of somatosensory feedback

and proprioception results in reduced balance confidence [44]. Unfortunately, no clear and consistent effect was observed across participants. The user preference was not clearly reflected in terms of gait biomechanics metrics, explored in this study. Although the user verbally expressed that gait stability was the factor in their reported reference, we did not observe consistent, large change of AP gait Margin of Stability for preferred control, compared to the other two control settings. This was potentially because the human's perceived walking instability is more sensitive than the measured balance metrics, as also observed in our previous study [45]. Another limitation was we only explored one stability index. Potential biomechanical variables that are more sensitive to balance instability than MOS should be considered, which could be a potential future work.

V. CONCLUSION

The study showed the feasibility of a novel hierarchical autotuning framework that can effectively and efficiently tune 12 robotic knee prosthesis control parameters to meet the user's preference while ensuring their safety. In addition, with the advantage of the utilized framework for selecting preferred prosthesis control, we showed that prosthetic knee users could consistently distinguish between different profiles tuned using the interface and displayed consistency in preference between these profiles. These results, although from a limited number of study participants, suggested that our framework has the potential to enable prosthesis users or clinicians to integrate the user's own perceptions and preferences into a prosthesis tuning procedure, simplified by the RL-based tuning algorithm. It should be noted that the preference studies include able bodied individuals, since the focus was on investigating if they can physiologically differentiate the different profiles. So, we cannot make any behavioral assertions on specific choices for the profiles. Furthermore, the preliminary gait biomechanical analysis showed that user preference was not clearly reflected in terms of the gait biomechanics metrics that we explored in this study. Due to the limited sample size (5 TF participants) and biomechanical metrics explored, additional biomechanical research is needed to understand the association of user preference with the gait performance measures.

Beyond the limitations discussed in the Discussion section, we also identified several other limitations and future work. Our experiments were conducted in a short timeframe. Further research is needed to answer the question about the consistency of preference over a longer timeframe. In addition, the explored gait performance measures were all only related to gait kinematics in this work. Computing the joint kinematics or monitoring EMG signals in the biomechanical analysis is needed in the future. Our study allowed changes only at control points that are fixed in gait phase for the participants to tune their prosthesis knee profile. In the future, we aim to provide participants a continuous knee feature space to explore when tuning the self-selected prosthesis control. Finally, one unexplored path that has the potential to further reduce the tuning time would be to investigate the sensitivity of users to different control points. Identifying the sensitivity to changes in each control point and how changes in one control point

affect perception of other control points could inform the design of future preference-based tuning such as the angular resolution and the number of control points. The successful investigation of such kind of sensitivity could further accelerate exploration and ensure people with amputation can personalize their prosthetic behavior with ease and efficiency.

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